

RESEARCH ARTICLE

A Design and Benchmarking Framework for Local Flexibility Markets in Active Power Systems

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ABSTRACT Local Flexibility Markets (LFMs) are emerging as a promising mechanism for utilizing Distributed Energy Resources (DERs) to support both transmission and distribution system operation. However, existing studies mainly focus either on conceptual discussions or on isolated market implementations, limiting the ability to systematically compare different architectures. This paper presents a unified design-decision framework for LFMs that, through a structured synthesis of the literature, organizes dominant market structures according to key operator choices, including flexibility scope, coordination mechanisms, bidding structures, and TSO–DSO interaction schemes. Based on this framework, representative LFM architectures are then quantitatively benchmarked under common modeling assumptions and shared network conditions. A combined transmission-distribution test system based on the IEEE 9-bus transmission network and multiple IEEE-33 distribution feeders is used for the assessment. Results demonstrate that the selected market architecture significantly affects congestion management capability, flexibility utilization, operational security, and total operating cost. In particular, architectures that neglect distribution network constraints may reduce operational costs but can lead to insecure DN operation, whereas coordinated sequential frameworks offer a more balanced trade-off between economic efficiency and secure network operation. The proposed framework provides a structured, implementation-oriented methodology for both organizing LFM design choices and benchmarking alternative architectures under consistent assumptions.

INDEX TERMS Local flexibility markets, TSO–DSO coordination, congestion management.

I. INTRODUCTION

The urgent need to mitigate climate change is accelerating the transition toward low-carbon power systems through the rapid integration of renewable energy sources and the electrification of end-use sectors such as transport, heating, and cooling [1], [2], [3]. While this transition is essential for decarbonization, it also introduces significant operational complexity across both transmission and distribution networks [4], [5]. The increasing deployment of electric vehicles, heat pumps, photovoltaics, and other distributed assets

alters traditional power flow patterns, increases uncertainty, and creates new coordination requirements [5], [6]. These developments can lead to technical challenges, including network congestion, voltage and thermal limit violations in distribution grids [7], [8], and increased balancing and flexibility needs at the transmission level [9]. At the same time, many electrified resources can provide valuable flexibility services [10], [11].

The Transmission System Operators (TSOs) and Distribution System Operators (DSOs) are required under the European Union's Regulation 2019/943 [12] to address grid-related issues through non-discriminatory, market-based mechanisms. In this context, ACER has developed

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complementary instruments covering both regulatory and technical dimensions. The Framework Guideline on Demand Response [13] defines the principles for non-discriminatory market access and participation of flexible resources, while the ACER Decision on the Flexibility Needs Assessment methodology [14] provides a harmonized approach for quantifying flexibility needs across transmission and distribution systems, specifying analytical procedures for TSOs and DSOs. In the same context, CEER [15] provides a comprehensive framework for how DSOs can procure flexibility through market-based mechanisms, including the necessary prerequisites, procedures, and roles of stakeholders.

Hence, Local Flexibility Markets (LFMs) have attracted increasing attention as a market-based means of unlocking DER flexibility for both distribution and transmission needs. Existing studies, however, are largely either conceptual or case-specific, which limits systematic comparison across alternative designs. This paper addresses this issue by organizing LFM architectures within a unified design-decision framework and by quantitatively benchmarking representative architectures under a common modeling setup.

A. LITERATURE REVIEW

According to [16] and [17], flexibility in power systems can serve different purposes depending on the system level: (i) flexibility for the transmission system, where flexibility is traditionally offered in wholesale markets by central/large-scale entities in the transmission system; (ii) flexibility for the transmission system provided by DERs located within the distribution network; and (iii) flexibility specifically for the distribution system, which is primarily utilized locally for voltage control, congestion management, and loss reduction in the distribution network.

In particular, [18] proposes a market-based framework for distribution-level flexibility procurement through coordinated local day-ahead and intra-day scheduling. Their design introduces a structured clearing process among DSOs, Balancing Responsible Parties (BRPs), and aggregator-suppliers, demonstrating that explicit coordination on the direction of flexibility is critical for achieving feasible and efficient market outcomes. However, the proposed structure limits the liquidity of flexibility, since wholesale markets cannot access DER flexibility. In [19], the authors propose a local day-ahead flexibility market formulation in which an aggregator schedules distributed resources to satisfy the DSO's upward and downward flexibility requests at minimum activation cost. However, unlike most LFM approaches, this formulation does not fully align with standardized market roles and product definitions typically adopted in real-world market designs. Moreover, the proposed methodology is not technology-agnostic, as the optimization model explicitly incorporates the detailed modeling and management of specific types of micro-flexible resources.

On the other hand, [20] proposes a near-real-time centralized flexibility market that considers both TSO and DSO constraints. Therefore, unlike previous approaches,

flexibility can be utilized by both TSOs and DSOs under a secure grid operation at minimum cost. However, the increased computational complexity and the lack of guarantees on global optimality remain important challenges for practical implementation and market transparency. Therefore, [21] proposes a sequential market framework in which the LFM is cleared before the balancing market, while only the remaining feasible flexibility is forwarded upstream. This approach provides a more realistic representation of system operation; however, ensuring that the forwarded flexibility does not violate distribution network constraints remains a key challenge.

To address this challenge, several smart forwarding and operational methods have been proposed. In [21], a computationally efficient order prequalification scheme is introduced to guarantee congestion-free operation of the distribution grid. Another approach is parametric optimization [22], in which the DSO iteratively solves the local market-clearing problem at different TSO–DSO exchange levels to determine a feasible flexibility range.

Moreover, several industrial and large-scale demonstration projects, such as EcoGrid 2.0 [23], SmartNet [24], PicloFlex [25], and EPEX [26], investigate the practical implementation of flexibility markets in real-world power systems. EcoGrid 2.0 [23] is a Danish project that tested a local flexibility market with 800 customers participating through electric heating units operating in parallel with the wholesale market. The project distinguishes between scheduled and conditional flexibility services, with the latter separated into reserve and activation components.

The SmartNet project [24] proposes a centralized LFM for real-time ancillary services that jointly considers transmission and distribution network constraints within a single optimization framework. Using detailed network models, the flexibility needs of TSOs and DSOs are implicitly determined to enable cost-optimal, secure activation of flexibility. Within the same project, alternative coordination schemes were also investigated, including local market formulations in which BRPs participate alongside system operators, thereby linking local flexibility procurement with broader market activities. EPEX SPOT [26] is a European power exchange that operates short-term electricity markets and acts as a platform for the collection and clearing of bids among flexibility service providers. The large-scale deployment of such platforms provides strong evidence of the effectiveness and practical viability of flexibility markets. For instance, in the United Kingdom, more than 100,000 assets participate in an LFM operated by EPEX SPOT on behalf of UK Power Networks DSOs, with approximately 2 GW of flexibility available for activation.

Despite the proposed individual approaches, a few review papers [16], [27] have been published that summarize the flexibility market designs and concerns. In [16], the authors review various local flexibility market approaches and focus on LFM formulations and their corresponding clearing methods. In contrast, [27] focuses on the non-technical

challenges associated with implementing LFMs, specifically governance issues, TSO–DSO coordination, inc–dec gaming, and competition/entry barriers. These challenges include role allocation and market transparency, coordination and data exchange between system operators, risks of strategic behavior across sequential markets, and limited competition that may reduce participation and market efficiency.

B. RESEARCH GAP AND PAPER CONTRIBUTIONS

Despite the growing body of work on LFMs, most studies remain either high-level and qualitative, focusing on definitions and conceptual comparisons without specifying how such markets should be structured in practice [16], [17], [27], [28], or architecture-specific, relying on case-dependent datasets and metrics that hinder transparent comparison across designs [18], [19], [20], [21], [29]. As a result, there is still a lack of a unified, decision-oriented organization of LFM architectures together with a consistent numerical benchmark under shared modeling assumptions.

To address this gap, this paper organizes existing LFM architectures within a unified framework by adopting common definitions and structuring them in a consistent sequence. The framework reflects the practical design choices a market operator must make when establishing an LFM, informed by system needs, regulatory constraints, and operational objectives. These choices shape the resulting market structure and explain the diversity of approaches proposed in the literature, including distribution-level parallel markets, centralized TSO–DSO co-optimization schemes, and sequential local-to-upstream clearing approaches. Based on this framework, representative architectures are quantitatively compared through a dedicated case study under a common modeling setup. In this context, the main contributions of this work are twofold:

- To provide a structured review and design-decision framework for LFMs, highlighting the key choices that market operators must define during market construction, including flexibility scope, coordination mechanisms, bidding structures, and TSO–DSO interaction schemes.
- To quantitatively evaluate representative market architectures under a common numerical setup and harmonized modeling assumptions, enabling a transparent comparison of their impact on congestion management, flexibility utilization, operational security, and total market cost.

II. BACKGROUND

A. KEY OPERATIONAL CHALLENGES

The ongoing transition toward high shares of DERs creates two major and interconnected classes of system-operation challenges: *distribution network (DN)* challenges and *transmission network (TN)* challenges. At the distribution level, networks originally designed for passive, one-way power delivery are increasingly operated as active, bidirectional systems due to the widespread integration

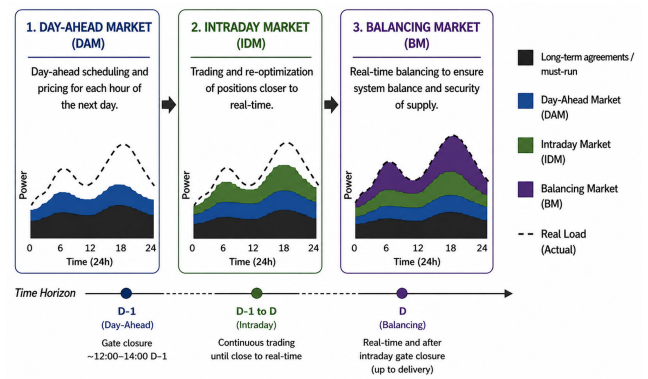


FIGURE 1. Conventional wholesale electricity market structure.

of DERs. This shift introduces local operational problems, such as reverse power flows, voltage violations, increased thermal congestion, increased line losses, and phase imbalance, making secure DN operation more complex for DSOs [16]. Consequently, distribution-level flexibility becomes essential to maintain technical limits and defer costly network reinforcements.

At the transmission level, the increasing share of intermittent DERs and the displacement of synchronous generation reduce available inertia and conventional frequency support, while increasing uncertainty in real-time balancing. As a result, TSOs face greater difficulty in maintaining power balance and frequency security, especially under significant disturbances [16]. In low-inertia operating conditions, events can produce higher RoCoF and deeper frequency nadirs, potentially triggering under-frequency load shedding (UFLS) and degrading reliability [4], [9]. In parallel, to satisfy minimum inertia and reserve requirements, TSOs may be forced to curtail RES in specific periods, reducing renewable utilization and overall system efficiency [30], [31].

B. CONVENTIONAL MARKET FRAMEWORK

Various wholesale market structures have been proposed in the literature and have already been implemented in real-world systems. For the purposes of this work, a simplified and practically implementable market structure is described, as illustrated in Fig. 1. The main operational concept is summarized as follows:

1) DAY-AHEAD MARKET (DAM)

The DAM constitutes the first stage of short-term electricity markets. In Europe, it is operated by power exchanges, whereas in the United States (U.S.), it is managed by Independent System Operators (ISOs) and Regional Transmission Organizations (RTOs). Its primary objective is to determine the optimal generation and consumption schedules for the following day, based on participants' bids. A key distinction between the European and U.S. day-ahead market designs lies in the way the market is structured. Europe uses the Single Day-Ahead Coupling (SDAC) model [32].

This is Europe's pan-European day-ahead market coupling mechanism, which clears zonal day-ahead auctions while simultaneously allocating scarce cross-zonal transmission capacity to maximize social welfare (economic surplus) [32].

2) INTRADAY MARKET (IDM)

The IDM, which follows the DAM, is a continuous wholesale energy market that allows market participants to adjust their positions closer to real time in response to improved forecasts and economic incentives. From an algorithmic perspective, the intraday market can be similar to the day-ahead market, with the main difference being that trades are executed closer to real time (typically up to 1 hour before delivery). As illustrated in Fig. 1, this represents a second step that adjusts previously cleared positions.

3) BALANCING MARKET (BM)

The BM constitutes the final operational stage before real-time system operation and is responsible for maintaining the continuous balance between generation and demand. In this market, TSOs procure and activate balancing reserves and flexibility services to compensate for forecast errors, unexpected disturbances, and real-time deviations from the schedules established in previous market stages. Depending on the system and market design, balancing services may include products such as Frequency Containment Reserve (FCR), automatic Frequency Restoration Reserve (aFRR), manual Frequency Restoration Reserve (mFRR), and Replacement Reserve (RR).

C. LOCAL FLEXIBILITY MARKET

Flexibility in power systems refers to the ability to adjust generation or consumption patterns in response to signals, such as prices or activation instructions, to provide different grid services in a cost-effective manner [16], [17]. However, an LFM is not completely different from conventional markets. The way the market is structured is crucial to avoid interactions between different markets that may create conflicts among market participants [24], where actions in one market can adversely affect outcomes in another. Such conflicts may arise when the same players participate in multiple markets, leading to situations in which activating flexibility in one market affects commitments or penalties in another. In addition, operational challenges arise when flexibility activation interacts with network constraints. Moreover, independent optimization of flexibility by TSOs and DSOs may lead to economically suboptimal solutions [20]. These three issues illustrate key concerns that should be considered in a market structure.

III. TSO-DSO FLEXIBILITY MARKET

In this section, the key design choices that the market operator must define during the market construction phase are presented as a structured decision sequence. For each design element, its definition, operational implications, advantages, and drawbacks are systematically discussed.

A. MARKET APPROACHES

The local markets can be categorized into three different types: i) exclusive DN flexibility, ii) exclusive TN flexibility, or iii) both TN and DN flexibility services.

1) EXCLUSIVE DN

This approach views LFMs as market-based mechanisms designed to enhance the operational capabilities of the DN. Within this framework, the market is operated locally, largely neglecting interactions with other network levels [18]. However, a key limitation of this architecture is that the procured flexibility remains confined to the local level and cannot be utilized by the TN or balancing market participants, since it is not forwarded to wholesale or balancing markets. Consequently, this may result in suboptimal system-wide operation and inefficient utilization of available flexibility resources. Moreover, certain DSO congestion management activations may operate in the opposite direction to the reserves required by the TSO, thereby forcing the TSO to activate additional balancing reserves to maintain system balance [33].

2) EXCLUSIVE TN

The second approach, as highlighted in the literature [24], integrates direct DER flexibility into the current market framework. However, it neglects DSO operational and contingency limits and fails to provide flexibility services to the DSO. Consequently, this may lead to suboptimal solutions and potentially insecure operation at the distribution level [16], [17], [33], as resources are primarily utilized to support TSO requirements [29]. Such an outcome is neither efficient nor desirable from a system-wide perspective.

3) DN-TN FLEXIBILITY

To overcome the aforementioned inefficiencies and ensure secure system operation, effective coordination between the DSO and the TSO is essential [20], [27]. This need has led to the development of a more comprehensive approach that integrates flexibility provision for both the DN and the TN. Such coordination can be achieved through various mechanisms, including co-optimized DSO-TSO flexibility markets [20] or the intelligent forwarding of residual flexibility from the DN to the TN, while strictly respecting DN operational constraints [21]. The sharing of flexible reserves between DSOs and TSOs can further improve the system's overall economic efficiency. However, these approaches typically involve complex coordination procedures.

B. MARKET PARTICIPANTS

As highlighted in [23] and [27], a key limitation of several market proposals is that the roles of different market participants are often not clearly defined or properly distinguished. We identify here the dominant market participants that may be involved in an LFM.

1) DISTRIBUTION SYSTEM OPERATOR (DSO)

DSOs are responsible for supplying electricity to consumers in an efficient, sustainable, and economically viable manner [16]. They must also ensure network security and maintain the reliability and quality of supply throughout the distribution system [17]. Within the LFM framework, DSOs are responsible either for providing the distribution network data and operational constraints required for market clearing and network feasibility assessment to a market coordinator [20], [21], [24], or for using this information internally to procure flexibility and issue flexibility requests [18], [29].

2) TRANSMISSION SYSTEM OPERATOR (TSO)

TSOs are responsible for the reliable and stable operation of the TN [17]. They ensure the real-time balance between electricity generation and demand at the system level, facilitate electricity market operation, coordinate cross-border power exchanges, and maintain system security under normal and contingency conditions. In addition, TSOs are responsible for congestion management, frequency control, and the procurement of balancing and ancillary services required for secure system operation.

3) BALANCE RESPONSIBLE PARTIES (BRPs)

The BRPs, usually wholesale suppliers or retailers, are entities tasked with maintaining the balance between supply and demand within their portfolio of producers and consumers, i.e., the net sum of injections and withdrawals over a specified time horizon. Participation in an LFM can benefit BRPs by enabling portfolio optimization, reducing imbalance volumes, and, consequently, lowering imbalance charges [17]. Some LFM structures [18] are designed in such a way that BRPs can actively participate, thereby increasing market competitiveness, for instance by acting in roles similar to a DSO and requesting flexibility.

4) AGGREGATORS

Individual prosumers possess limited flexibility volumes, which reduces their negotiation power in LFMs. To effectively utilize this distributed flexibility potential, aggregators consolidate the small-scale resources of multiple prosumers into aggregated flexibility services that can be traded in LFMs [16], [17]. In practice, the aggregator is often represented by the same entity as the supplier. This assumption is primarily motivated by the need to avoid conflicting incentives between separate market actors [33]. For instance, if these roles are assigned to different entities, an aggregator's activation of flexibility may alter end-users' actual consumption, thereby deviating from the energy schedule previously submitted by the supplier in the wholesale market. This deviation leads to imbalances for which the supplier is financially liable. As a result, the aggregator may generate revenues from flexibility provision, while the supplier incurs imbalance penalties, leading to a misalignment between control actions and economic responsibility. By unifying the

roles of aggregator and supplier, these conflicts are mitigated, as a single entity jointly optimizes energy scheduling and flexibility [18].

5) MARKET OPERATOR

The market operator should provide a flexible trading platform and is also responsible for market clearing. The debate around LFM usually centers on whether the role of market operator should be allocated to the DSO or to an independent operator [27]. As some LFM structures assign the market operator role to the DSO, concerns arise under the legal unbundling framework when the DSO is a part of a vertically integrated company.

In such cases, two major risks can emerge. First, conflicts of interest may arise when the DSO has incentives to favor affiliated market participants (e.g., suppliers or aggregators within the same corporate group). Second, information asymmetry becomes a critical issue, as the DSO has privileged access to detailed network data (e.g., congestion locations, flexibility needs) that could be exploited to benefit specific actors [27]. These concerns motivate considering independent market operators to ensure neutrality, transparency, and a level playing field among participants.

C. MARKET PRODUCT

Defining market products is a fundamental task in the design of LFMs, as it determines how flexibility is specified, traded, and ultimately utilized. A market product corresponds to a set of rules and attributes that define the service being procured. These definitions must account for system requirements, while remaining technology-neutral to ensure fair participation across different types of flexibility resources. Importantly, not all product attributes are treated equally within the market framework. Some attributes are directly embedded in the bidding process, while others are enforced as prequalification or technical requirements.

Technically, a flexibility activation is characterized by a power modification (increase or decrease) that occurs at a defined time, lasts for a specified duration, and occurs at a particular node within the distribution network. These attributes characterize the fundamental nature of the product and define what is being traded in the market. As such, flexibility can be fully described using the following five attributes [16], [28]:

- 1) **Direction:** Indicates whether the activation increases or decreases power (upward or downward).
- 2) **Power Modulation:** The magnitude of power change that the flexibility can deliver.
- 3) **Starting time and trigger:** The initiation time of the activation and the signal or event that triggers it.
- 4) **Duration:** The time period over which the flexibility is maintained.
- 5) **Location:** The specific node or location within the distribution system where the flexibility is provided.

Additional attributes are also frequently considered [17], [24], although they are not always explicitly included in the

market-clearing process. These may include the rate of change, response time, availability window, ramping constraints, the maximum number of activations over a given time horizon, minimum and maximum activation durations, minimum time intervals between consecutive activations, and limits on the cumulative energy that can be injected or withdrawn over a specified period.

D. BIDDING AND MARKET CLEARING MECHANISMS

Another key decision in market design concerns the choice of bidding structure. Different bidding architectures can significantly affect the LFM's operation. They influence not only how aggregators behave and design their internal optimization algorithms, but also how the market-clearing algorithm is formulated and implemented. Bids are not aggregated across nodes; instead, market participants can submit multiple bids for different nodes [24].

1) PTU-BASED INDEPENDENT CLEARING

The most widely used bidding approach is the bid-per-programmable-time-unit (PTU) framework, in which bids are submitted and cleared independently for each time interval. In this classical formulation, the market-clearing process determines the accepted flexibility for each PTU without explicitly accounting for intertemporal dependencies. This allows DSOs and BRPs to procure the exact amount of flexibility required at each time step, providing high temporal granularity and operational simplicity. However, this independence across PTUs introduces significant limitations for aggregators. Most flexibility services offered by aggregators rely on load shifting, which inherently couples decisions across time. This may lead to infeasible or suboptimal schedules for aggregators and can result in unintended deviations in total energy consumption [18].

2) PTU-BASED CLEARING WITH INTER-TEMPORAL CONSTRAINTS

To address the above limitations, practical implementations extend the PTU-based framework by allowing market participants to submit bids that include inter-temporal constraints spanning multiple PTUs [18]. These constraints are typically defined at the bid level as explained previously in Section III-C. Under this enhanced formulation, while the market is still organized around discrete PTUs, the clearing process is no longer fully independent across time. This enables aggregators to submit physically consistent flexibility strategies and reduces the risk of infeasible partial activations. In [24] a specific subcategory of such strategy is defined as *All-or-None*, meaning that during the clearing process submitted offers are either fully accepted or entirely rejected.

Subsequently, there can be three main types of PTU bids for each time interval, as illustrated in Fig. 2. The first is the non-curtailable bid, where the bidder submits a fixed quantity (Q , in MW) and a corresponding price (P , in EUR/MWh) for each time step. In this case, the accepted quantity cannot be adjusted by the market-clearing algorithm, so the bid is

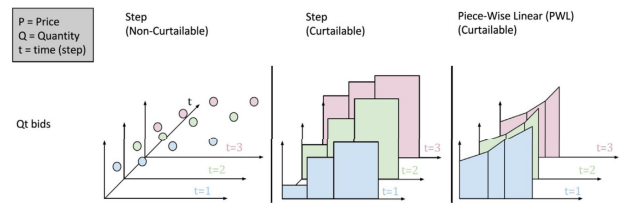


FIGURE 2. Different types of standard bids [24].

either fully accepted or fully rejected. Depending on the bidding structure (independent or inter-temporal constraint), acceptance may apply to each individual time step or to the entire time horizon. The second type is the curtailable bid, in which a single price ($P_0 = P_1$) is defined over a range of quantities ($Q_0 - Q_1$), allowing partial acceptance within that range. The third type is the piecewise-linear bid, which is also curtailable and whose price varies with quantity, enabling a more flexible representation of marginal costs.

In any case, despite the price bidding structure, all necessary market products should be included in the bidding process to align with the market standards. After the market-clearing process, various pricing mechanisms can be used to determine the remuneration for accepted bids. The two most common approaches are pay-as-bid and pay-as-clear.

- 1) Pay-as-bid: In the pay-as-bid scheme, each accepted participant is paid according to its bid price, encouraging strategic bidding [19]. This means that market participants are not incentivized to bid their real costs, potentially introducing an economic distortion [24].
- 2) Pay-as-clear: In contrast, in the pay-as-clear scheme, all accepted bids are remunerated at a uniform market-clearing price, typically determined by the marginal accepted offer.

E. MARKET TIMING DESIGN PARAMETERS

Timing parameters play a critical role in market design, as they shape the procedural structure and sequencing of market operations. The most important timing-related parameters are:

- **Time Horizon:** Defines the period over which market clearing is performed. LFM can operate across different horizons, including day-ahead, intraday, and real-time markets.
- **Time Step:** Specifies the temporal granularity of the market and determines the resolution at which decisions are made. The time step cannot exceed the selected time horizon.
- **Gate Closure Time (GCT):** Defines the deadline after which no new bids or modifications to existing orders can be submitted.
- **Maximum Market-Clearing Duration:** Specifies the maximum computational time available to the market operator for clearing the market and obtaining a feasible solution.

F. OPTIMIZATION MODELS AND OBJECTIVE FUNCTION

A review of the existing works reveals that the objective functions adopted in market-clearing algorithms predominantly converge to two main formulations: activation cost minimization and social welfare maximization [16]. The selection between these objectives is not independent, as it inherently constrains and influences several of the preceding design choices within the market framework.

1) MINIMIZE ACTIVATION COST

Models that minimize activation cost are typically system-operator-centric, as their primary objective is to reduce the cost of flexibility activation and other regulatory actions ($\min \sum C|\Delta P|$). In such formulations, the requestors of flexibility, namely, TSOs and/or DSOs, do not participate in the market through explicit bids. Instead, they provide their forecast network state, which implicitly reflects their flexibility needs. Consequently, within this optimization framework, System Operators (SOs) are assumed to behave as price takers [24]. Moreover, a pay-as-clear pricing mechanism cannot be directly applied, as no explicit marginal price is defined within this cost-minimization structure. Nevertheless, pricing signals can be derived through alternative methods, such as post-processing techniques or by identifying the marginal accepted offer.

2) MAXIMIZE SOCIAL WELFARE

Social welfare is defined as the sum of the benefits of all participants in the market [16]. This type of market optimization is widely adopted in practice because it relies on a social-welfare-maximization framework that combines supply and demand curves. In such models, both bids and offers are explicitly submitted, enabling a transparent and competitive market environment. Compared to alternative formulations, this approach is generally more tractable, allows a broader set of market actors (e.g., BRPs) to participate, and offers greater transparency and simplicity, thereby enhancing its practical applicability. According to the European Balancing Guideline (EBGL) [34], the maximization of social welfare is the recommended objective for market clearing.

Formulating the optimization problem is central to validating and assessing the quality of market outcomes. It emerges from the sequence of prior design choices, which are often influenced by considerations of computational tractability. Ensuring transparency and non-discriminatory outcomes requires a mathematically sound formulation that delivers fair solutions. However, in LFMs, integrating detailed network constraints is not always feasible due to the NP-hard nature of the OPF problem. This challenge is less prominent in wholesale markets, where simplified formulations such as DC-OPF are commonly used and considered acceptable due to their computational efficiency and because DC approximations are generally suitable for high-voltage network physics.

IV. DESIGN FRAMEWORK FOR LOCAL FLEXIBILITY MARKETS

The design of an LFM is not a single process but rather a combination of interdependent design choices. As discussed in Section III, in practice, this corresponds to selecting a specific configuration from the set of alternatives presented. Different combinations of these choices lead to fundamentally different market architectures, with varying levels of complexity, coordination requirements, operational feasibility, and market outcomes. It is challenging to provide a comprehensive classification of all LFM models, given the wide variety of designs proposed in the literature. In this work, instead, we construct dominant representative LFM configurations, as proposed in the literature, by combining different design choices.

A. EXPLICIT COST-BASED LOCAL DISTRIBUTION NETWORK MARKET

The *Cost-based LM model* can operate in all time frames. For instance, it can run in DA and ID local markets [18], or BM [20]. In any case, for proper operation across different markets, the output (final setpoints) of one market is used as input to the next market [21]. For instance, a wholesale DA market cannot be cleared if the local day-ahead market does not finalize the aggregator base profile. An example of an explicit DN market approach is given in (1). The objective function minimizes the regulatory cost of all resources (1a). The constraints include the flexibility bidding limits (1b) and the feasible operating region of the DN (1c).

$$\text{Cost-based LM} : \begin{cases} \min F(\Delta P) = \sum C |\Delta P| & (1a) \\ \text{s.t. } P^- \leq \Delta P \leq P^+, & (1b) \\ x^{DS} \in \mathcal{X}^{DS} & (1c) \end{cases}$$

In this formulation, ΔP denotes the flexibility activation, defined as the deviation from the reference schedule established in the previous market stage or from baseline operation. The coefficient C denotes the activation cost of flexibility and, for simplicity, is assumed to be identical for both upward and downward flexibility requests. Nevertheless, the formulation can be readily extended to accommodate asymmetric activation prices. Finally, $\mathcal{X}^{DS}$ represents the feasible operating region of the distribution network, comprising all operating points that satisfy the distribution-system operational constraints, including voltage limits, thermal loading limits, and power-flow feasibility conditions.

B. EXPLICIT FLEXIBILITY TRANSMISSION NETWORK MARKET

The *TN-FM* operates similarly to the current wholesale market, with additional flexible resources integrated in aggregated form to enhance transmission network operation. As a result, the DSO is not required to provide any network model or constraints. An example of this market structure is given in (2). The objective function maximizes the social

welfare, as shown in (2a), similar to the current structure. The constraints include the flexibility bidding limits (2b) and the feasible operating region of the transmission network (2c).

$$TN - FM : \begin{cases} \max \sum_{b \in \mathcal{B}} \lambda_b^{bid} P_b - \sum_{o \in \mathcal{O}} \lambda_o^{offer} P_o & (2a) \\ \text{s.t. } P^- \leq P_o \leq P^+, & (2b) \\ x^{TS} \in \mathcal{X}^{TS}. & (2c) \end{cases}$$

where, P_o denotes the flexibility quantity offered by market participants, while λ_o^{offer} represents the corresponding offer price. Similarly, P_b and λ_b^{bid} denote the requested flexibility quantity and the associated willingness-to-pay of the market participants submitting bids. The objective function maximizes social welfare by maximizing the difference between the value of accepted bids and the cost of accepted offers. Finally, $\mathcal{X}^{TS}$ represents the feasible operating region of the transmission system that satisfies the adopted transmission-network constraints, such as power balance equations and transmission line capacity limits.

C. SEQUENTIAL DN AND TN FLEXIBILITY

Algorithm 1 summarizes the sequential market-clearing framework, which links the independently cleared local and transmission flexibility markets. First, the DSO clears the explicit local flexibility market by solving the *cost-based LM* in (1). The flexibility procured by the DSO is then deducted from the corresponding original offers, thereby determining the residual flexibility available for transmission-level services and preventing it from being allocated simultaneously to both system operators. The residual flexibility is subsequently forwarded to the TSO, where two alternative forwarding strategies are considered ($k \in \{A, B\}$).

In *Case A*, the forwarding process is non-intelligent, i.e., each offer is simply reduced by the flexibility already activated by the DSO before being forwarded to the TSO. In contrast, *Case B* incorporates a network-aware forwarding procedure inspired by merit-order-based TSO–DSO validation schemes, similar to those proposed in [35], although alternative validation approaches, such as [21], can be readily adopted. The residual offers are first sorted in ascending order of offer price so that lower-cost flexibility is preserved whenever curtailment is required. If the forwarded flexibility violates any distribution network constraints, the validation procedure progressively reduces the highest-priced offer group until all voltage, thermal-loading, and power-flow constraints are satisfied. When multiple offers share the same clearing price, they are proportionally reduced using a common scaling factor to preserve non-discriminatory treatment among economically equivalent participants. Finally, the validated forwarding limits replace the original offer limits, and the TSO clears the *TN-FM* according to (2).

The corresponding pseudocode is given in Algorithm 1. For notational clarity, each flexible resource $i \in \mathcal{F}$ is associated with offer bounds ($\underline{P}_i, \bar{P}_i$) and price c_i (corresponding

Algorithm 1 Sequential DN and TN Procedure

Require: Original flexibility offers $\mathcal{O} = \{(\underline{P}_i, \bar{P}_i, c_i)\}_{i \in \mathcal{F}}$; forwarding case $k \in \{A, B\}$.

- 1: Solve the *Cost-based LM* in (1)
- 2: Obtain DSO activation ΔP_i^{DSO}

Ensure: Forwarded flexibility set $\mathcal{O}_k^F$.

- 3: **for all** $i \in \mathcal{F}$ **do**
- 4: Compute the residual flexibility limits:

$$\underline{P}_i = \underline{P}_i - \Delta P_i^{DSO}, \quad \bar{P}_i = \bar{P}_i - \Delta P_i^{DSO}.$$
- 5: **end for**
- 6: **if** $k = A$ **then**
- 7: Set $\underline{P}_{i,A}^F \leftarrow \underline{P}_i$ and $\bar{P}_{i,A}^F \leftarrow \bar{P}_i$ for all $i \in \mathcal{F}$.
- 8: **else**
- 9: Initialize $\underline{P}_{i,B}^F \leftarrow \underline{P}_i$ and $\bar{P}_{i,B}^F \leftarrow \bar{P}_i$.
- 10: Construct the merit-order list by sorting the residual offers in ascending order of c_i .
- 11: Evaluate the distribution-network operating point, assuming activation of the forwarded flexibility.
- 12: **while** the DSO constraints are violated **do**
- 13: Identify the highest-price active offer group $\mathcal{G} \subseteq \mathcal{F}$.
- 14: **if** $|\mathcal{G}| = 1$ **then**
- 15: Reduce the forwarded limits of the offer $i \in \mathcal{G}$ until either:
 (i) the DSO constraints are satisfied,
 or
 (ii) the offer is fully removed.
- 16: **else**
- 17: Determine a common reduction factor $\rho \in [0, 1]$ for all $i \in \mathcal{G}$.
- 18: Update

$$\underline{P}_{i,B}^F \leftarrow \rho \underline{P}_{i,B}^F, \quad \bar{P}_{i,B}^F \leftarrow \rho \bar{P}_{i,B}^F, \quad i \in \mathcal{G}.$$
- 19: **end if**
- 20: Re-evaluate the distribution-network constraints.
- 21: **end while**
- 22: **end if**
- 23: Forward $\mathcal{O}_k^F$ to the TSO.
- 24: The TSO clears the *TN-FM* using (2), with the original offer limits replaced by the forwarded limits $\underline{P}_{i,k}^F$ and $\bar{P}_{i,k}^F$.

to P^- , P^+ , and C in (1)–(2)); ΔP_i^{DSO} denotes the DSO activation, and $\underline{P}_{i,k}^F, \bar{P}_{i,k}^F$ are the forwarded limits under case k . In Case B, $\mathcal{G}$ and ρ denote the highest-price active offer group and its common reduction factor, respectively.

V. CASE STUDY

In the case study considered, the aforementioned market architectures are implemented and comparatively evaluated. For a fair comparison between the TSO's social-welfare maximization and the DSO's cost minimization, the TSO is assumed to be a price-taker, allowing both formulations to be expressed as cost-minimization problems. In addition

to the flexibility-based market architectures, two baseline wholesale market cases without flexibility are considered for benchmarking. The first corresponds to a conventional copperplate wholesale market formulation without TN constraints (*Copperplate WM*), while the second incorporates transmission network constraints through a DC-OPF formulation (*DC-OPF WM*). The proposed framework is generic with respect to the market horizon and can therefore be applied to different scheduling and operational timescales.

An important aspect of all market structures is the baseline operation, which can significantly affect market behavior [36]. However, in our analysis, this is treated as constant to enable comparing design frameworks. For all architectures that account for network constraints, the transmission network is modeled using DC power-flow formulations, while the distribution network is represented using AC-based formulations. In this work, a second-order cone programming (SOCP) relaxation is adopted for the distribution network, and its exactness is verified through a post-processing validation phase. The optimization problem is implemented in the Pyomo framework and solved using the Gurobi solver.

A. MARKET STRUCTURE OPTIONS

Following the market structure presented in Section III, BRPs are omitted for simplicity. A pay-as-bid market-clearing mechanism is considered to capture the different revenues arising from different market structures, since pay-as-clear mechanisms generally require a larger pool of offers to adequately reflect their price-formation characteristics. Moreover, a PTU-based independent clearing process is adopted, where flexibility bids can be activated anywhere between their minimum and maximum values at the same offered price. Inter-temporal bidding constraints are not considered, as they primarily affect the formulation of FSP bids rather than the proposed network-aware market-clearing framework. Since this work focuses on the network-side methodology and DSO-TSO coordination, PTUs are treated independently.

B. INPUT DATA AND NETWORK

The IEEE 9-bus system is adopted as the TN, comprising three conventional generation units and one wind power unit, as depicted in Fig. 3. Part of the aggregated demand of the three load buses is represented through IEEE-33 medium-voltage distribution feeders. For instance, *Feeder A* corresponds to 5% of the total demand connected to Bus 5. The load profile at Bus 7 is more industrial/commercial-oriented, with high midday consumption and lower demand in the afternoon–evening period, whereas Buses 5 and 9 follow more residential-based patterns with an evening peak; among the two, Bus 9 is more stressed than Bus 5. For the flexibility-enabled case studies, five energy storage systems are integrated into the distribution network at

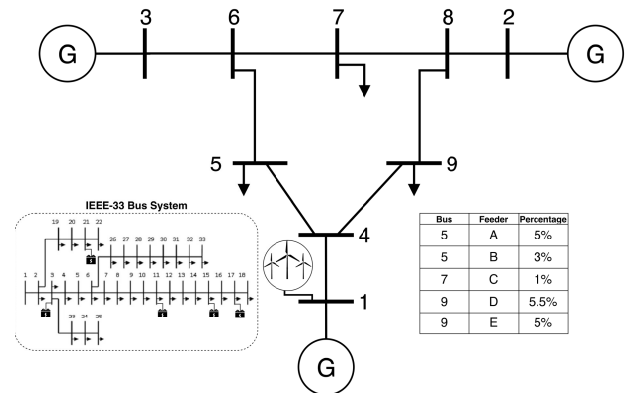


FIGURE 3. IEEE 9-bus TN with embedded IEEE 33-bus DN feeders at Buses 5, 7, and 9, and a wind park at Bus 4.

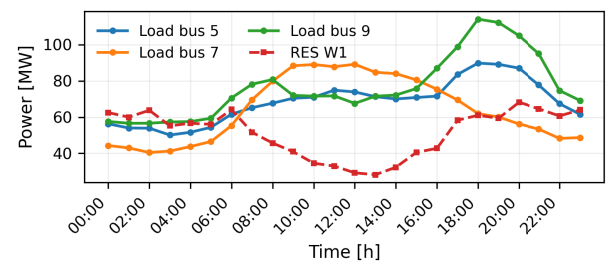


FIGURE 4. Demand and RES generation profiles considered in the simulations.

TABLE 1. Conventional generation offers.

Unit	Bus	$P_g^{\min}$ (MW)	$P_g^{\max}$ (MW)	Offer price (€/MWh)
G_1	1	10	250	20
G_2	2	10	300	30
G_3	3	10	270	45

buses [3, 11, 15, 17, 21]. The aggregated load profiles and the forecasted wind park generation are presented in Fig. 4.

For the conventional generation units, the submitted offers are provided in Table 1. The offer prices are selected as representative values to establish a realistic merit order among the generators. Since the objective of this work is to evaluate the interaction between the transmission and distribution flexibility markets rather than reproduce a specific wholesale electricity market, the exact bid values are not critical to the proposed methodology. Regarding the flexibility support from the battery systems, aggregated discharging offers of 4 MW are assumed for both TN Buses 7 and 9 during the period 16:00–21:00, with an offer price of 15 €/MWh. The batteries are assumed to be co-located with PV parks and to be fully charged by 16:00. For the purposes of this analysis, their flexibility is offered at a sufficiently low price relative to conventional generators, allowing the submitted bids to be fully committed if required. Accordingly, the inter-temporal energy constraint is represented by limiting the total activation over the considered period to the available stored energy. In practice, aggregators would account for inter-temporal dependencies and the risk of non-delivery when determining their flexibility offers through their internal bidding strategies.

TABLE 2. Market operation comparison.

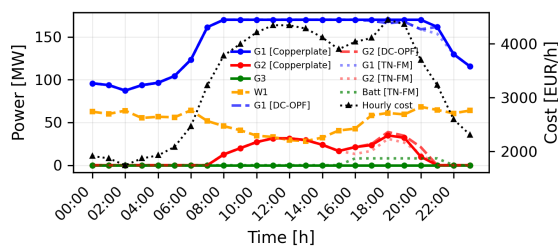
Architecture	Wholesale Cost [€]	LFM Cost [€]	Total Cost [€]	Flex. Util.	#DN Violations Voltage/Line	#TN Viol.
Copperplate WM	78,870	–	78,870	–	Feeder D: 6/3, Feeder E: 4/0	4
DC-OPF WM	79,056	–	79,056	–	Feeder D: 6/3, Feeder E: 4/0	0
TN-FM	78,416	–	78,416	100%	Feeder C: 6/0	0
Cost-based LM	78,975	41	79,016	11%	0	0
Sequential FM - Case A	78,375	41	78,416	100%	Feeder C: 6/0	0
Sequential FM - Case B	78,400	41	78,441	96%	0	0

VI. RESULTS AND DISCUSSION

The objective of this section is to assess how the different flexibility coordination schemes influence the overall system operation, congestion management capability, flexibility utilization, and market costs.

A. GENERATION AND FLEXIBILITY DISPATCH

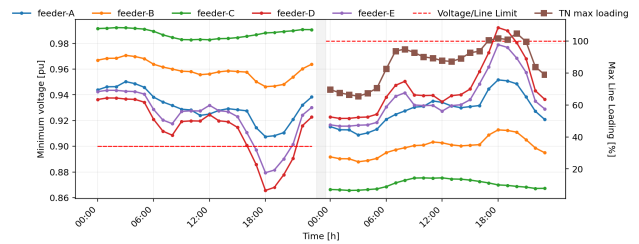
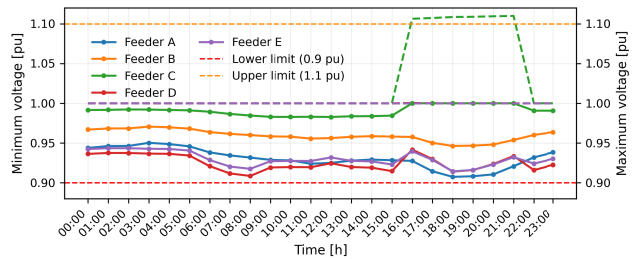
For the *Copperplate WM* results, TN constraints are neglected; therefore, the market-clearing outcome is determined by the merit order of the submitted generation offers. The resulting dispatch is illustrated in Fig. 5, where the cheapest resources, namely the Wind Park and G1, operate at their maximum available power, while the next cheapest unit (G2) is dispatched to supply the remaining demand. In contrast, the *DC-OPF WM* incorporates TN congestion constraints into the market-clearing process. In the *Copperplate WM* case, both G1 and the Wind Park are located in the same area, which results in line loading violations. Consequently, as shown with dashed lines in Fig. 5, the *DC-OPF WM* requires the redispatch of conventional generation to alleviate congestion, thereby increasing G2 utilization and reducing G1 dispatch.

**FIGURE 5. Dispatch of units for the *Copperplate WM*, *DC-OPF WM*, and *TN-FM* models.**

When flexibility resources are integrated, the battery energy storage system, which represents the lowest-cost flexibility resource after the Wind Park, is utilized as much as possible during the offered flexibility window (see the dotted lines in Fig. 5). Similarly, the Wind Park operates at its maximum available power due to its low operational cost.

B. CONGESTION AND COST ANALYSIS

The *Copperplate WM* model represents the most economical operation among the cases without flexibility support. However, as shown in Table 2 and Fig. 6, it also corresponds to the

**FIGURE 6. Minimum DN voltage results for each feeder and TN maximum line loading.****FIGURE 7. DN feeder minimum and maximum voltage results under the *TN-FM* and *Sequential FM Case A*.**

most insecure operating condition, where both transmission and distribution network violations occur. More specifically, Feeders D and E experience voltage violations, with voltages dropping below 0.9 p.u., while several distribution lines exceed the 100% loading limit. Feeder D presents 6 voltage-violation time steps within a single day, along with 3 thermal-loading violations on DN lines. In comparison, Feeder E shows 4 voltage violations but none for thermal loading. At the transmission level, four line-loading violations are also identified on line 1–4. Therefore, the *DC-OPF WM* avoids those TN violations through redispatching, with a small increase in the total cost due to the commitment of the more expensive generator G2.

When flexibility services are introduced, part of the congestion management burden is shifted toward distributed flexible resources, thereby reducing the need for costly redispatch actions. In particular, the *TN-FM* scenario achieves the lowest total operating cost; however, DN violations still occur since DN constraints are not considered in the market-clearing process. More specifically, as shown in Fig. 7, in Feeder C (industrial/commercial-oriented), the high reverse power flow caused by the BESS located within the DN during evening hours creates a new congestion issue that was not observed in the cases without flexibility resources. At the same time, the previously observed voltage violations are eliminated due to the increased BESS power injection during the critical violation hours within the DN.

In contrast, the *Cost-based LM*, through a dedicated distribution network market, maintains fully secure network operation by leveraging available flexibility to mitigate voltage and line congestion at the distribution level. Nevertheless, the total operating cost is higher, as the flexibility is dedicated

exclusively to distribution network needs and is therefore unavailable for transmission-level optimization.

The *Sequential FM Case A*, which does not employ an intelligent forwarding technique, produces results similar to those of the *TN-FM*. This is because the remaining flexibility after the *Cost-based LM* is forwarded to the transmission network and, when activated by the wholesale market, distribution network violations reappear. The main difference between the two architectures lies in their cost allocation. In the sequential structure, flexibility procured for distribution-level needs is remunerated through the LFM, while flexibility activated for TN support is settled through the wholesale market. This creates two separate cost streams and provides a fairer allocation of flexibility costs according to their use, as reflected in columns 2–3 of Table 2.

Finally, the *Sequential FM Case B*, which incorporates a “smart” forwarding technique, appears to provide the best trade-off between operational cost and secure network operation. More specifically, only the flexibility that preserves secure DN operation is forwarded to the transmission network. Therefore, regardless of how the remaining wholesale flexibility is activated, safe grid operation is maintained. Across all local flexibility architectures, the LFM cost remains identical, as they address the same voltage and line congestion issues shown in Fig. 6. Therefore, the DSO has no incentive to procure additional flexibility, as it would provide no further congestion relief.

C. FLEXIBILITY UTILIZATION ANALYSIS

Table 2 presents the utilization levels of the available flexibility resources under the different market architectures. Given the low offer price of the storage systems, a 100% utilization level corresponds to accepting all submitted flexibility bids. Among the four flexibility-based architectures, the *Cost-based LM* primarily activates flexibility from resources in congested feeder areas, resulting in only 11% utilization. Consequently, a significant portion of the available flexibility remains unused. The *TN-FM* and *Sequential FM Case A* fully utilize the available flexibility potential; however, this results in overvoltage violations in Feeder C due to high battery discharge power during periods of relatively low demand. Finally, the *Sequential FM Case B* appears to provide the most appropriate operating outcome, utilizing 96% of the available flexibility while maintaining all network constraints within acceptable limits, thanks to the smart forwarding technique.

D. DISCUSSION

The obtained results demonstrate that explicit flexibility procurement can substantially reduce congestion management costs while improving network security. However, the efficiency of flexibility utilization strongly depends on the selected market architecture and the coordination mechanism between system operators. The exclusive DN and TN approaches improve the operation of their respective network layers but may lead to suboptimal overall system utilization

due to the lack of coordination between local and system-wide objectives. The sequential TSO–DSO coordination framework achieves the best overall performance among the investigated architectures. This behavior is mainly attributed to the progressive filtering and forwarding of feasible flexibility resources, which allows the TSO optimization problem to operate only on flexibility bids that preserve the secure operation of the distribution network.

VII. CONCLUSION

This paper presented a unified design framework and comparative evaluation of representative LFM architectures proposed in the literature. The analysis showed that the selected market structure and the coordination mechanism between TSOs and DSOs strongly affect congestion management capability, operational security, flexibility utilization, and total system cost. The results demonstrated that architectures that neglect DN constraints may achieve lower operating costs but can lead to insecure network operation, whereas coordinated sequential approaches provide a more balanced trade-off between flexibility utilization and secure system operation. Furthermore, intelligent flexibility-forwarding was shown to significantly improve interaction between local and wholesale flexibility markets.

Future work will build on these findings by extending DSO flexibility procurement beyond congestion relief to account for additional distribution-network benefits, including network losses and asset aging. This is expected to increase the value and utilization of local flexibility, which remains limited when procurement is driven solely by voltage and thermal congestion. Building on this extended formulation, smart flexibility-forwarding methodologies will be investigated to enable the practical implementation of sequential LFM architectures. Particular attention will be given to preserving DN security while maximizing the flexibility available to upstream markets, as results indicate that sequential coordination offers the most promising trade-off between operational security and economic efficiency.

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